

Exam Portal - Project Summary

Exam Portal is a full-stack web application for creating and taking online quizzes/exams. It is built with a Spring Boot (Java) backend and a React frontend, and supports two user roles: Admin and User.

Tech Stack

Layer	Technology
Backend	Spring Boot 2.7.3, Spring Security, Spring Data JPA, JWT (jjwt)
Database	MySQL (via mysql-connector-java)
Frontend	React 18, Redux, Axios, Bootstrap 5
Auth	JWT-based stateless authentication with role-based access (Admin/User)

Backend Architecture

Package: com.project.examportalbackend - organized into standard Spring Boot layers:

- configurations - JwtRequestFilter, JwtUtil, SecurityConfig (JWT filter chain & security rules)
- controllers - AuthController, CategoryController, QuestionController, QuizController, QuizResultController
- models - User, Role, Category, Quiz, Question, QuizResult, LoginRequest/Response
- repository - Spring Data JPA repositories for each model
- services / services.implementation - business logic layer (Auth, Category, Question, Quiz, QuizResult, UserDetails)

Frontend Architecture

React app (Create React App) structured with actions/reducers/store (Redux), services (API calls via Axios), and pages split by role:

- pages/admin - categories, questions, quizzes management (add/update/delete)
- pages/users - quiz browsing, attempting quizzes, and viewing report cards
- components - shared/reusable UI pieces
- Login and Registration flows integrated with JWT auth

Core Features

- Admin: create/update/delete quiz categories, quizzes, and questions
- Admin: view and manage all quizzes and their questions
- User: register/login, browse quizzes by category, attempt quizzes
- User: view profile and past quiz report cards/scorecards
- Role-based access control secured via JWT tokens

Notes

The repository also includes a Project-Synopsis.pdf with additional project documentation, and a set of UI screenshots (images/) covering login, registration, admin management pages, and user quiz-taking flows.